NITHIN THIRUMANI

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Professional Summary

Software Engineer with **4+ years of combined software engineering and graduate research experience** developing **Python automation tools, backend applications, automated tests, and reliable software systems**. Experienced in **test automation, software quality assurance, debugging, API validation, root cause analysis**, and application performance optimization. Strong programming foundation in **Python, C#, Java**, and object-oriented design, with hands-on experience using **Pytest, Vitest, Git, Jenkins, GitHub Actions, Linux**, and CI/CD workflows. Skilled in translating technical requirements into testable software functionality, identifying defects, validating system behavior, and maintaining reusable development tools. Graduate research experience includes **AI-driven workflows, model evaluation, and automated validation**, with an interest in developing reliable diagnostic tools and software-hardware integration systems.

Technical Skills

Programming — Python, C#, Java, JavaScript, TypeScript, SQL, C, Object-Oriented Programming, Data Structures, Algorithms
Backend — RESTful APIs, FastAPI, API Integration, JSON, CRUD Operations, Pydantic, Microservices, Modular Software Design
Testing & Quality — Test Automation, Pytest, Vitest, Unit Testing, Integration Testing, Regression Testing, API Testing, Test Case Design, Debugging, Defect Investigation, Root Cause Analysis, Software Validation
AI & Data — Generative AI, Large Language Models, AI-Assisted Development, Machine Learning, Model Evaluation, Automated Validation, Data Processing
Cloud & DevOps — GitHub Actions, Jenkins, CI/CD, Linux, Unix Command Line, Docker, Kubernetes, AWS, Azure, Cloudflare, Railway, Terraform
Databases & Tools — Git, GitHub, GitHub Projects, Jira, Slack, PostgreSQL, MySQL, SQL Server, MongoDB, GitHub Copilot

Professional Experience

Texas A&M University–Corpus Christi, Corpus Christi, TX — Mar 2025 – Present
Graduate Research Assistant

- Develop reusable **Python-based automation tools** for AI research applications, implementing structured data preparation, model execution, output validation, and result-processing workflows to support reliable experimental testing.
- Design and implement modular **software testing and validation components** to verify model outputs, identify unexpected behavior, and evaluate software functionality across different research configurations.
- Build automated **experimental execution and evaluation workflows** using Python, organizing input preparation, execution steps, result collection, and validation procedures to improve repeatability and consistency.
- Investigate **software failures and unexpected execution behavior** through systematic debugging, performance analysis, and comparison of experimental results to identify potential causes and evaluate corrective approaches.
- Develop and evaluate **Machine Learning, Deep Learning, and Large Language Model (LLM)** solutions, implementing model inference, benchmarking, and output validation workflows for research applications.
- Analyze **software performance and processing bottlenecks** by benchmarking execution workflows, inspecting computational behavior, and testing alternative configurations to improve efficiency and reliability.
- Optimize **Python data-processing and inference workflows** through input preparation improvements, execution adjustments, and reusable processing components to support consistent experimental results.
- Implement structured **testing procedures and edge-case validation** to identify inconsistent outputs, investigate unexpected behavior, and verify correctness across different experimental conditions.
- Collaborate with faculty and research teams to **understand technical requirements**, discuss implementation approaches, investigate software issues, and refine research applications through technical feedback.
- Prepare and maintain **technical documentation, testing procedures, experimental records, and evaluation reports**, documenting software behavior, validation methods, implementation decisions, and findings to support reproducibility.

AD IT Robotics, India — May 2022 – Nov 2024
Software Development Engineer - Backend

- Developed and maintained **Python-based backend software** for an automobile spare parts inventory management system, implementing workflows for SKU management, stock tracking, purchasing and sales records, supplier information, inventory adjustments, and low-stock monitoring.

- Engineered **RESTful APIs and CRUD operations** using **Python, SQL, and JSON**, incorporating object-oriented programming, input validation, and exception handling to maintain predictable application behavior.
- Designed and implemented a **dynamic multi-constraint inventory filtering feature** using Python and SQL, supporting simultaneous searches by SKU, product name, vehicle model, manufacturer, supplier, price range, stock availability, and product category.
- Diagnosed **application defects and performance issues** by reproducing slow inventory searches, testing different filter combinations, inspecting generated SQL queries, and measuring execution times to identify inefficient operations.
- Optimized **SQL query performance** by refining `WHERE` predicates, adding indexes on frequently searched columns, and implementing dynamic query conditions to improve database retrieval efficiency.
- Improved **backend application responsiveness** by reducing unnecessary database calls, consolidating SQL operations, and implementing pagination to limit retrieved records during complex inventory searches.
- Developed reusable **Python automation scripts** for repetitive inventory updates, bulk record processing, reporting, and reconciliation, reducing manual operations and supporting consistent data-processing workflows.
- Implemented **backend transaction processing and validation logic** for stock adjustments, purchasing and sales records, and inventory updates, incorporating exception handling to manage invalid inputs and unexpected conditions.
- Performed **unit, integration, regression, and API testing** to validate backend functionality, reproduce software defects, investigate failures, verify data accuracy, and confirm expected behavior following implementation changes.
- Applied **Git, Jenkins, Linux, and CI/CD practices** during development, testing, debugging, and maintenance, supporting version-controlled software changes, repeatable development workflows, and reliable application delivery.

Projects

Jarvis - An Intelligent Personal Finance Assistant
JavaScript, Cloudflare Workers, Cloudflare D1, REST APIs, PWA, Web Push, GitHub Actions

- Developed a **cloud-based personal finance application** using **JavaScript, Cloudflare Workers, and Cloudflare D1**, implementing backend workflows for income, expenses, debts, credit cards, recurring bills, savings, and natural-language financial inputs.
- Designed and validated a deterministic **financial forecasting and payment prioritization engine** with account-aware cash-flow calculations, funding-gap detection, protected savings, and scheduled reminders; incorporated **automated testing** to verify financial rules and interconnected transaction workflows.

ApplyWise - Evidence-Grounded Job Application Assistant
Python, FastAPI, React, PostgreSQL, Pytest, Vitest, GitHub Actions, Docker

- Co-developed a **full-stack web application** using **React, FastAPI, and PostgreSQL**, implementing modular workflows for career evidence management, job-description analysis, generated-statement validation, and approval tracking; tested unsupported claims and invalid evidence references using a simulated AI provider.
- Served as **Project Manager and developer** for a six-member team, coordinating test requirements, development tasks, GitHub Issues, pull-request reviews, and **GitHub Actions CI**; the team developed **230 automated tests** using **Pytest and Vitest** and recorded 85 commits at its October 2026 checkpoint.

Intelligent Data Quality and Enrichment Pipeline
Python, Pydantic, PostgreSQL, AWS, Pytest, Git, GitHub Copilot

- Developed a configurable **Python data-processing and validation pipeline** using **Pydantic and PostgreSQL**, implementing structured record validation, schema consistency checks, and reusable transformations to identify missing, incomplete, and inconsistent data.
- Implemented reusable **validation components, exception handling, audit logging, and automated unit tests** to verify processing behavior, investigate invalid records, and improve reliability across data enrichment workflows.

Microservices-Based Healthcare Monitoring System
Python, FastAPI, React.js, MongoDB, REST APIs, Docker, Kubernetes

- Developed a **microservices-based healthcare monitoring application** using **Python, FastAPI, and MongoDB**, implementing four independent backend services for patient records, simulated real-time vital readings, threshold-based alerts, and consolidated health reports.
- Integrated **RESTful APIs** with a React dashboard and a Python simulator generating heart rate, blood pressure, and temperature readings; containerized services with **Docker** and deployed them using **Kubernetes**, gaining experience with software integration and simulated device-data workflows.

Certifications

AWS Certified Cloud Practitioner
HashiCorp Certified: Terraform Associate
Microsoft Certified: Azure AI Engineer Associate

Education

Texas A&M University–Corpus Christi, Corpus Christi, TX — Jan 2025 – Dec 2026
Master of Science in Computer Science — GPA: 3.6/4.0
JB Institute of Engineering & Technology, Hyderabad, India — Dec 2020 – May 2024
Bachelor of Computer Science — Cumulative GPA: 8.14/10.00